FINAL PROJECT REPORT

Food Ordering Behaviour and Consumer Trends

Date	27 August 2026
Team ID	-
Project Name	Food Ordering Behaviour and Consumer Trends
Maximum Marks	5 Marks

1. Introduction

1.1. Project Overview

The Food Ordering Behaviour and Consumer Trends project focuses on analyzing customer food-ordering patterns using Tableau. The project uses a structured food-ordering dataset containing approximately 50,000 order records and 19 attributes. The dataset provides information related to customers, food preferences, restaurants, meal types, order values, discounts, delivery, ratings, repeat orders, mood, hunger level, weather conditions, and other order-related factors. Tableau is used to transform the raw data into meaningful visualizations, dashboards, filters, and stories.

1.2. Objectives

- To analyze food-ordering behaviour and consumer trends.
- To identify cuisine preferences across different cities.
- To analyze order distribution across age groups and restaurant segments.
- To identify customer preferences across different cuisines.
- To analyze orders across companies and meal types.
- To study customer ratings across different meal types.
- To identify cities with higher order volumes.
- To compare average order values across restaurant types.
- To analyze the relationship between discounts and repeat orders.
- To develop interactive Tableau visualizations and dashboards.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Food-ordering platforms generate large amounts of customer and order-related data. Raw tabular data makes it difficult to identify customer preferences, ordering patterns, spending behaviour, and differences between cities, cuisines, restaurants, and meal types. The project addresses the need to analyze this food-ordering data and present meaningful insights through Tableau visualizations and dashboards.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a Tableau-based data analytics and visualization system. The raw Food Ordering Behaviour dataset is imported into Tableau, checked for quality, prepared for analysis, and transformed into visualizations. The implementation follows: Raw Dataset → Data Quality Check → Data Preparation → Data Analysis → Visualization → Dashboard → Story → Insights.

Key Features:

- Cuisine preference analysis across cities.
- Age-group and restaurant-segment analysis.
- Cuisine-wise customer preference analysis.
- Company and meal-type analysis.
- Customer rating analysis.
- City-wise order-volume analysis.
- Restaurant category and average order-value analysis.
- Discount and repeat-order analysis.
- Interactive Tableau filters.
- Dashboard and story presentation.

2.3. Initial Project Planning

Phase 1 – Requirement Analysis	Understand project objective, dataset, and required problem statements.
Phase 2 – Data Collection	Obtain and import the Food Ordering Behaviour dataset in CSV format.
Phase 3 – Data Preparation	Check data quality, data types, missing values, duplicates, and inconsistencies.
Phase 4 – Data Exploration	Explore customer, cuisine, restaurant, meal, order, and behavioural fields.
Phase 5 – Visualization Development	Create charts and visualizations for the defined business questions.
Phase 6 – Dashboard Development	Combine relevant visualizations into an interactive Tableau dashboard.
Phase 7 – Story Development	Organize visualization views into a logical data story.
Phase 8 – Testing and Documentation	Test filters and visualizations and document findings and outcomes.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Project Overview	The project analyzes food-ordering behaviour and consumer trends using the provided Food Ordering Behaviour dataset.
Data Collection Plan	The data is collected from the provided Food Ordering Behaviour dataset in CSV format and imported into Tableau for quality checking, preparation, analysis, and visualization.
Raw Data Source	Food Ordering Behaviour Dataset; approximately 50,000 records and 19 attributes; CSV format; used for customer, order, restaurant, cuisine, meal, rating, discount, and repeat-order analysis.

3.2. Data Quality Report

Data Quality Issue	Severity	Resolution Plan
Missing or null values	Medium	Check required fields for null values and handle them appropriately before analysis.
Duplicate order records	Medium	Check Order Id values and remove duplicate records if identified.
Incorrect data types	Medium	Verify dimensions and measures and

		assign appropriate Tableau data types.
Inconsistent categorical values	Low	Check and standardize categorical values such as City, Cuisine, Meal Type, and Restaurant Type where required.
Unusual numerical values	Low	Validate numerical fields such as Order Value, Delivery Fee, Rating Given, and Time Taken to Order.

3.3. Data Exploration and Preprocessing

Data Overview	Approximately 50,000 food-order records and 19 attributes covering customer, order, restaurant, delivery, and behavioural information.
Data Cleaning	Checked missing values, duplicate records, inconsistent entries, and field types before analysis.
Data Transformation	Applied filtering, sorting, grouping, and aggregation required for the Tableau analyses. COUNT(Order Id) and AVG(Order Value) were used where appropriate.
Data Type Conversion	Verified numerical fields such as Order Value, Delivery Fee, Rating Given, and Time Taken to Order, and categorical dimensions such as City, Cuisine, Meal Type, and Restaurant Type.
Column Splitting and Merging	No major column splitting or merging was required because the required information was already available in separate columns.
Data Modeling	The project uses a single CSV dataset, so relationships between multiple tables were not required.
Save Processed Data	The prepared Tableau data source/workbook was saved for visualization, dashboard, story, and documentation development.

4. Data Visualization

4.1. Framing Business Questions

Business Question	Visualization	Fields Used	Purpose
1. What are the different food preferences across cities?	Heat Map	City, Cuisine, Order Id	Compare food and cuisine preferences across cities.
2. How are orders distributed across different age groups and restaurant segments?	Bar Chart	Age Group, Restaurant Type, Order Id	Understand order distribution among age groups and restaurant categories.
3. What is the distribution of customer preferences across different cuisines?	Bar Chart	Cuisine, Order Id	Identify the most and least preferred cuisines.
4. How do orders vary across different companies and meal types?	Grouped Bar Chart	Company, Meal Type, Order Id	Compare ordering patterns across companies and meal categories.
5. How do customer ratings vary across different meal types?	Pie Chart	Meal Type, Rating Given	Compare customer ratings across meal types.
6. Which cities generate the highest number of food orders?	Bar Chart	City, Order Id	Identify cities with higher and lower order volumes.
7. Which restaurant type has the highest average order value?	Horizontal Bar Chart	Restaurant Type, Order Value	Compare customer spending across restaurant categories.
8. Is there a relationship between discounts and repeat orders?	Stacked Bar Chart	Discount Applied, Is Repeat Order, Order Id	Examine the relationship between discount availability and repeat-order behaviour.

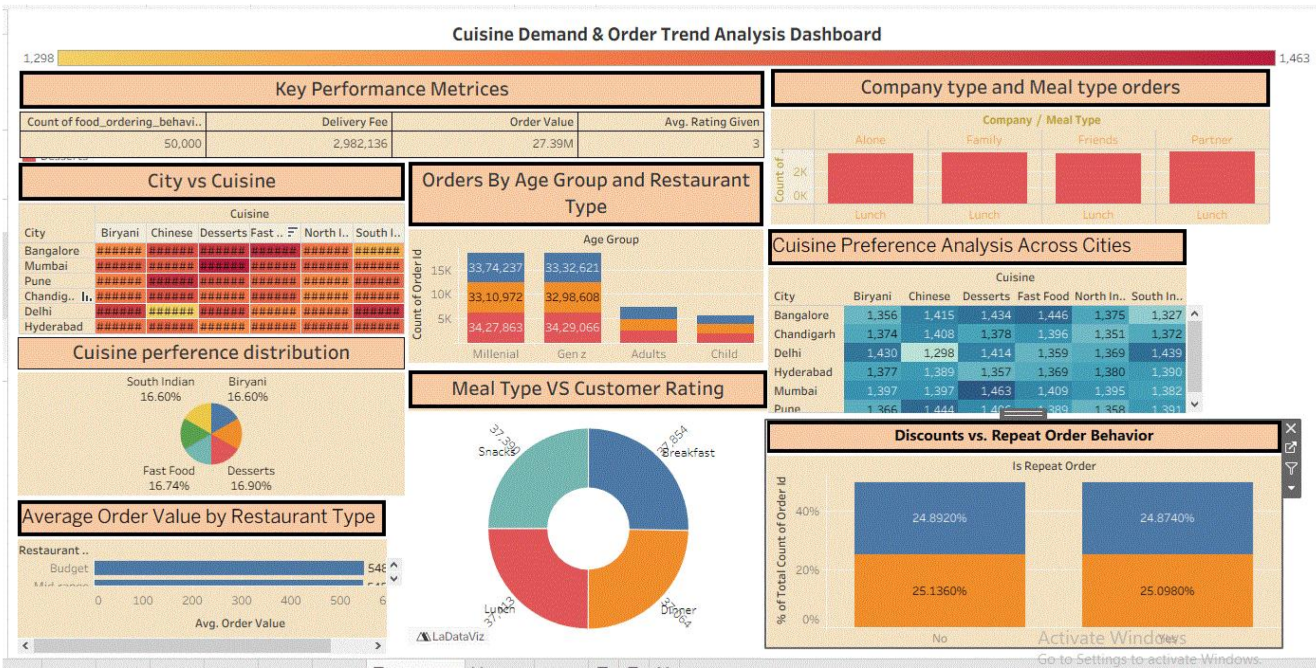
4.2. Developing Visualizations

The visualizations were developed in Tableau using suitable dimensions, measures, aggregations, and filters. The project uses heat maps, bar charts, grouped bar charts, pie charts, horizontal bar charts, and stacked bar charts to present the eight business analyses.

5. Dashboard

5.1. Dashboard Design File

The dashboard combines relevant Tableau visualizations into a consolidated interactive view. The design uses clear titles, suitable chart types, readable layouts, and interactive filters to support exploration of food-ordering behaviour. Major dashboard areas include cuisine preferences, city-wise orders, restaurant categories, average order value, meal types, customer ratings, company-wise ordering, and discount/repeat-order behaviour. Interactive filters include City, Cuisine, Meal Type, and Restaurant Type, where applicable.

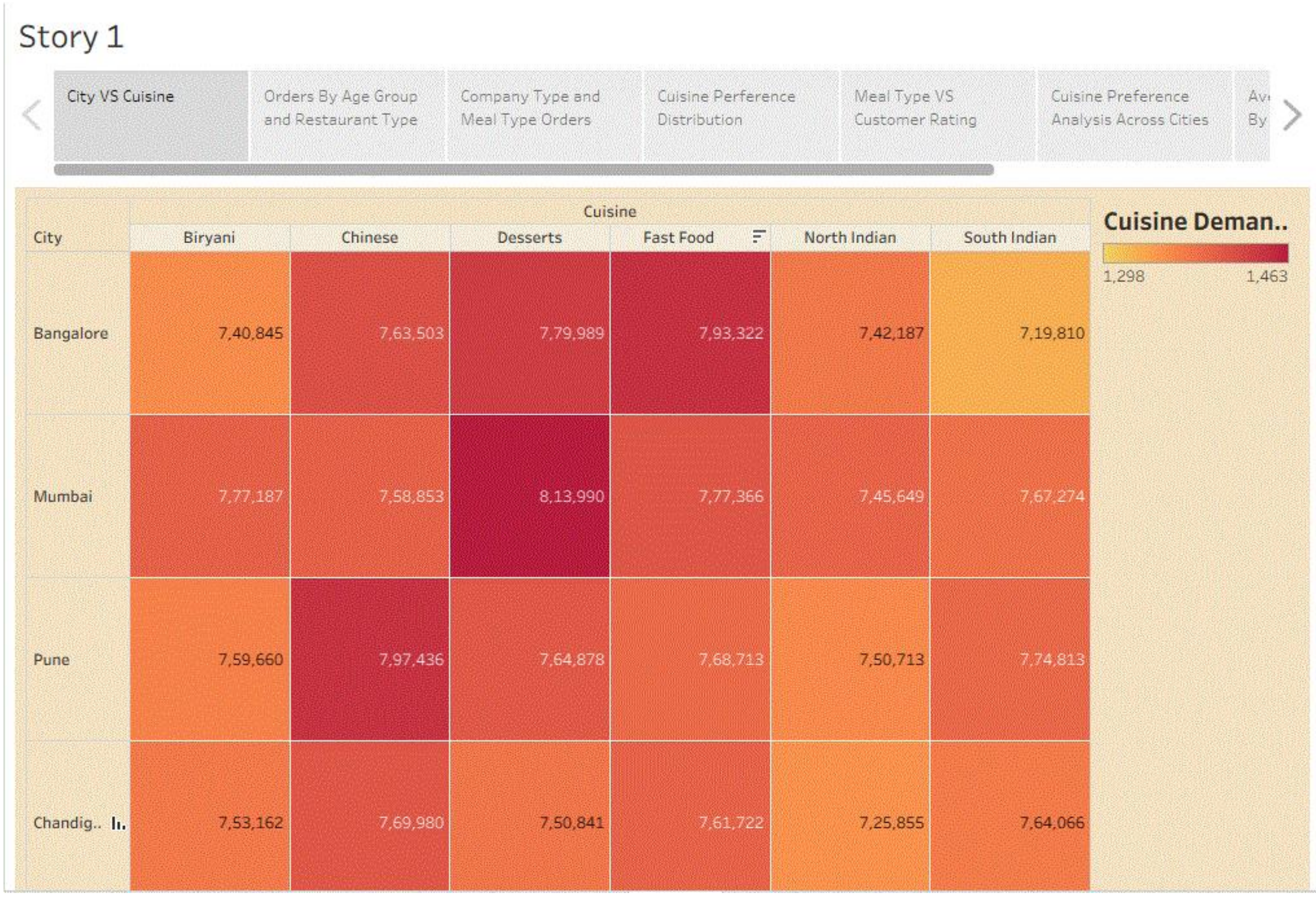


6. Report

6.1. Story Design File

The Tableau Story organizes the project analysis into a logical sequence so that the audience can understand the findings from food preferences and customer behaviour through restaurant analysis, order volume, ratings, spending patterns, and repeat-order behaviour.

Story flow: Food Preferences → Customer Behaviour → Restaurant Analysis → Order Volume → Customer Ratings → Spending Analysis → Repeat Orders.



7. Performance Testing

7.1. Utilization of Data filters

Interactive data filters were implemented in the Tableau dashboard to enable users to explore specific portions of the dataset efficiently. The primary filters used in the project include **City, Cuisine, Meal Type, and Restaurant Type**. These filters allow users to dynamically analyze customer preferences, order patterns, and cuisine demand across different categories.

7.2. No of Calculation Field

Sr.no.	**Calculated field**	**Purpose**
1	Age group	Categorize with customer into different age groups for analysis

7.3. No of Visualization

The project contains **8 primary business-question visualizations** designed to analyze food-ordering behaviour and consumer trends. These visualizations include:

- **1 Heat Map**
- **4 Bar/Horizontal Bar Charts**
- **1 Grouped Bar Chart**
- **1 Pie Chart**
- **1 Stacked Bar Chart**

These visualizations provide insights into **city-wise cuisine demand, meal preferences, restaurant types, customer ratings, and overall consumer behaviour**. Interactive filters further enhance the usability of the dashboard by allowing users to explore the data based on different dimensions.

8. Conclusion/Observation

The Food Ordering Behaviour and Consumer Trends project demonstrates how Tableau can transform food-ordering data into meaningful visual insights. The analysis of approximately 50,000 food orders provides insights into cuisine preferences, city-wise order volumes, restaurant categories, meal types, customer ratings, spending behaviour, and repeat-order patterns.

- Cuisine preferences differ across cities.
- Order distribution varies across age groups and restaurant segments.
- Customer preferences differ across cuisines.
- Ordering patterns vary across companies and meal types.
- Customer ratings can be compared across meal types.
- Order volumes differ across cities.
- Average order value can be compared across restaurant types.
- Discount and repeat-order analysis helps examine customer behaviour.
- Interactive filters make the dashboard easier to explore.

9. Future Scope

- Integrate a live database or regularly refreshed data source instead of relying only on a static CSV.
- Add advanced customer segmentation based on ordering frequency, spending, and preferences.
- Introduce predictive analysis for order-volume or customer-behaviour forecasting.
- Develop a recommendation component for cuisines or meal types.
- Extend geographic analysis with more detailed map-based views.
- Automate data refresh and dashboard updates.
- Add advanced dashboard interactivity such as parameters and drill-down features.
- Extend the dashboard for continuous monitoring of order, satisfaction, and customer-behaviour metrics.

10. Appendix

10.1. Source Code(if any)

This project is primarily implemented using Tableau. No separate programming source code is required for the core visualization implementation. The main project artifact is the Tableau workbook/data source containing the prepared data, worksheets, visualizations, dashboard, filters, and story.

10.2. GitHub & Project Demo Link

GitHub Repository	https://github.com/diptishete122/Food-Ordering-Behaviour-and-Consumer-Trends/upload/main/DA%20with%20tableau%20report%20template
Project Demo Link (Tableau Public)	https://public.tableau.com/app/profile/dipti.shete/viz/Foodorderingdashboard_17878350585100/Dashboard1
Dashboard Link	https://public.tableau.com/app/profile/dipti.shete/viz/Foodorderingdashboard_17878350585100/Dashboard1
Story Link	https://public.tableau.com/app/profile/dipti.shete/viz/Foodorderingstory_17878353765530/Story1